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# The herbal medicine inchin-ko-to (TJ-135) induces apoptosis in cultured rat hepatic stellate cells

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**Abstract**

Use of herbal remedies in the treatment of various diseases has a long tradition in Eastern medicine and the liver diseases are not an exception. In their use, lack of elucidation of mechanism(s) as well as randomized, placebo-controlled clinical trials has been a problem. Recently, we and others reported that inchin-ko-to (TJ-135), one of herbal remedies, suppressed hepatic fibrosis in animal models. In the course of clarifying the mechanism, we directed our focus on hepatic stellate cells (HSCs), playing a pivotal role in hepatic fibrosis, and found that rat HSCs cultured with TJ-135 changed their morphology to star-like configuration with thin, slender and dendritic processes with fewer stress fibers, which might be the features in apoptosis. In fact, TJ-135 induced HSC apoptosis in a time- and concentration-dependent manner as judged by the nuclear morphology, quantitation of cytoplasmic histone-associated DNA oligonucleosome fragments and caspase 3 activity. In HSCs treated with TJ-135, increased expression of p53 and decreased expression of Bcl-2 and phosphorylated Akt and Bad were determined. HSC apoptosis is shown to be involved in the mechanisms of spontaneous resolution of rat hepatic fibrosis and the agent which induces HSC apoptosis has been shown to reduce experimental hepatic fibrosis in rats. Thus, the induction of HSC apoptosis could be the mechanism how TJ-135 works on the resolution of hepatic fibrosis. Our current data may shed light on the novel effect of the herbal remedy.


*Keywords:* Hepatic fibrosis; Akt; Bad

## Introduction

Use of herbal remedies in the treatment of various diseases has a long tradition in Eastern medicine and the liver diseases are not an exception (Schuppan et al., 1999). Although interferon and other anti-viral agents have been applied to viral hepatitis, which is the most prevalent liver disease in the Eastern world, elimination of hepatitis virus is not always achieved and their therapeutic efficacy is limited. In this situation, herbal remedies are widely used especially in Japan and China to reduce the liver damage and/or fibrosis for the patients of viral hepatitis who have been unresponsive to those anti-viral agents.

In the use of herbal remedies, lack of elucidation of mechanism(s) on their effect as well as randomized, placebo-controlled clinical trials has been a problem (Schuppan et al., 1999). Regarding those used for the liver diseases, the evidence supporting their therapeutical efficacy in animal experiments, especially as anti-fibrotic agents, has been accumulating, first as to sho-saiko-to (Shimizu et al., 1999; Sakaida et al., 1998) and then recently inchin-ko-to (TJ-135) (Imanishi et al., 2004; Inao et al., 2004; Sakaida et al., 2003). TJ-135 has been reported to suppress liver injury in mice

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induced by Fas (Yamamoto et al., 2000) and hepatic fibrosis in rats induced by a choline-deficient L-amino acid-defined diet (Sakaida et al., 2003), thioacetamide (Imanishi et al., 2004) or repeated injections of carbon tetrachloride or pig serum (Inao et al., 2004). Moreover, the beneficial effect of TJ-135 on postoperative biliary atresia was reported; the improvement of markers of hepatic fibrosis such as hyaluronic acid, prolyl hydroxylase, procollagen III peptide and type VI collagen was determined (Kobayashi et al., 2001). Regarding the mechanism(s) how TJ-135 would suppress those liver injury, it has been shown that TJ-135 prevents apoptosis of cultured rat hepatocytes (Yamamoto et al., 1996, 2000) and suppresses proliferation and fibrogenesis of cultured rat hepatic stellate cells (HSCs) (Imanishi et al., 2004; Inao et al., 2004) and those of human hepatic stellate cell line (Sakaida et al., 2003). In this study, we directed our focus on the TJ-135 effect on HSCs in culture to further examine the mechanism(s) and found that TJ-135 causes interesting morphological changes in HSCs.

## Materials and methods

### *Animals*

Male Sprague–Dawley rats (Shizuoka Laboratory Animal Center, Shizuoka, Japan) were fed a standard pelleted diet and water ad libitum and used in all the experiments. All animals received humane care in compliance with the institution's guidelines.

### *Materials*

Anti-Bcl-2, anti-Akt and anti-phospho-Akt was obtained from BD Biosciences, anti-p53 (Pab240), anti-Bax (N-20) anti-Bad (C-20) and anti-glyceraldehyde-3-phosphate dehydrogenase (GAPDH) (V-18) from Santa Cruz Biotechnology, anti-phospho-Bad from Cell Signaling Technology and anti-α-smooth muscle actin from BioMakor. Inchin-ko-to (TJ-135) was kindly provided by Tsumura and Co., Tokyo, Japan, and was directly dissolved in the culture medium with 10% fetal calf serum.

### *Cell isolation and culture*

HSCs were isolated from the rats, weighing 300–400 g, using a metrizamide (Sigma Chemical Co., St. Louis, MO) gradient centrifugation, as previously described (Ikeda and Fujiwara, 1995). The isolated cells were seeded on uncoated plastic tissue-culture dishes (Falcon, Lincoln Park, NJ) at a starting density of $1-4\times10^5$ cells/cm$^2$ and cultured in Dulbecco's Modified Eagle's Medium (Nissui Pharmaceutical Co., Ltd., Tokyo, Japan) containing 10% fetal calf serum (GIBCO, Grand Island, NY). The confluent cells were subcultured as previously described (Pinzani et al., 1989), and experiments were performed between 2 and 3 weeks after isolation (between 1 and 2 passages). In these cells, no differences in response to TJ-135 were found irrespective of times of passage. All the experiments were conducted in the presence of 10% fetal calf serum.

### *F-actin staining*

HSCs were plated on Lab Tek slides (4 wells; Nalge Nunc International, Naperville , IL). After treatment, F-actin staining was performed as previously described (Yanase et al., 2000) and examined by fluorescence microscopy.

### *Hoechst staining*

HSCs were plated on Lab Tek slides (4 wells). After treatment, Hoechst staining was performed as previously described (Leroy-Viard et al., 1995) and examined by fluorescence microscopy.

### *TdT-mediated dUTP nick-end labelling (TUNEL) staining*

HSCs were plated on Lab Tek slides (4 wells). After treatment, apoptotic cells were detected by an in situ cell death detection kit (Roche Applied Science) with TUNEL technique using the manufacturer's instructions.

### *Cell death detection enzyme-linked immunosorbent assay*

A cell death detection enzyme-linked immunosorbent assay kit (Roche Diagnostics, Indianapolis, IN) was used to quantitatively determine cytoplasmic histone-associated DNA oligonucleosome fragments associated with apoptotic cell death. Briefly, after treatment, HSCs were lysed with 200 μl of lysis buffer and incubated for 30 min at room temperature. Then, 20 μl of supernatant was transferred into the streptavidin-coated microtiter plate, and 80 μl of the immunoreagent was added to each well. After incubation at room temperature for 2 h, the solution was decanted, and each well was rinsed three times with incubation buffer. Color development was carried out by adding 100 μl of 2,2′-azino-di(3-ethylbenzthiazoline sulfonate) (ABTS) solution, and absorbency was measured at 405 nm against ABTS solution as a blank.

### *Caspase 3 activity assay*

HSCs cultured in 100-mm-diameter dishes were harvested and pelleted by centrifugation. The medium supernatant was discarded and the cell pellet was washed in 1 ml of ice-cooled phosphate-buffered saline. Caspase 3 (DEVDase) activity was determined by a colorimetric CaspACE kit (Promega, Madison, WI) using the manufacturer's instructions.

### *Immunoblot analysis*

After experimental treatment, the medium was discarded, and HSCs were incubated in 25 mM HEPES, pH 7.5, 150 mM NaCl, 1% Igepal CA-630, 10 mM $MgCl_2$, 1 mM EDTA and 10% glycerol at 4 °C. After clearing lysates of insoluble cell debris by centrifugation, samples containing the same amount

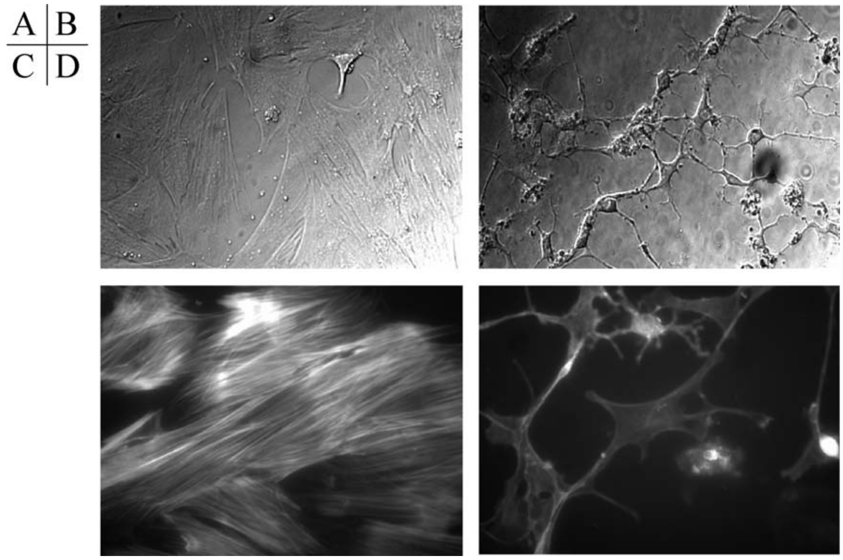


Fig. 1. Effect of TJ-135 on cellular morphology and actin stress fibers in HSCs. HSCs were cultured with (B and D) or without (A and C) TJ-135 at 400 μg/ml for 72 h. Cellular morphology was judged by light microscopy (A and B, ×400), actin stress fibers by fluorescent staining with phalloidin (C and D, ×600). Representative results of six experiments are shown.

of protein were separated by sodium dodecyl sulfate–polyacrylamide gel electrophoresis under reducing conditions, which were transferred to a sheet of polyvinylidene difluoride membrane (Amersham). To block nonspecific binding, the membrane was soaked in blocking agent derived from skim milk (Blockace; Snow Brand Milk Product Co., Ltd., Sapporo, Japan) for 1 h at room temperature. Then, it was incubated with primary antibody (dilution 1:1000; antibodies used: α-smooth muscle actin, p53, Bcl-2, Bax, Akt, phospho-Akt, Bad, GAPDH and dilution 1:2000, antibodies used: phospho-Bad) overnight at 4 °C. The membrane was incubated with horseradish peroxidase-conjugated second antibody (dilution 1:1000) for 1 h at room temperature. Immunoreactive proteins were visualized using a chemiluminescence kit (Amersham) and recorded in a chemiluminescence recording system (LAS 1000; Fuji-film, Tokyo, Japan).

*Statistical analysis*

When indicated, statistical analysis was performed by Student's *t*-test, and $P<0.05$ was considered significant.

## Results

*Effect of TJ-135 on HSC morphology and cytoskeletal organization*

HSCs used in this study were "activated" with a myofibroblast-like phenotype in the process of culture on plastic with enhanced proliferation, fibrogenesis and contractility (Rockey et al., 1992; Friedman et al., 1989). The addition of TJ-135 at 400 μg/ml to culture-activated rat HSCs resulted in morphologic alteration of shrinking within 24 h as judged by light microscopy. As demonstrated in Fig. 1, HSCs changed from a flattened fibroblastic phenotype (Fig. 1A) with abundant stress fibers assessed by F-actin staining (Fig. 1C) to star-like configuration with thin, slender and dendritic processes (Fig. 1B) with fewer stress fibers (Fig. 1D) by the treatment with 400 μg/ml TJ-135 for 72 h. These data suggest that TJ-135 altered morphology and cytoskeletal organization by inducing a dynamic reorganization of actin filaments in HSCs.

Among actin, increased expression of α-smooth muscle actin is one of the key features in activation of HSCs (Rockey et al., 1992). As previously reported (Inao et al., 2004), TJ-135 reduced the expression of α-smooth muscle actin; incubation of HSCs with 400 μg/ml TJ-135 caused the decrease in the expression of α-smooth muscle actin as early as 12 h. TJ-135 at the lower concentration, 100 μg/ml, also reduced the expression of α-smooth muscle actin at 48 h from its addition to the culture medium of HSCs as depicted in Fig. 2.

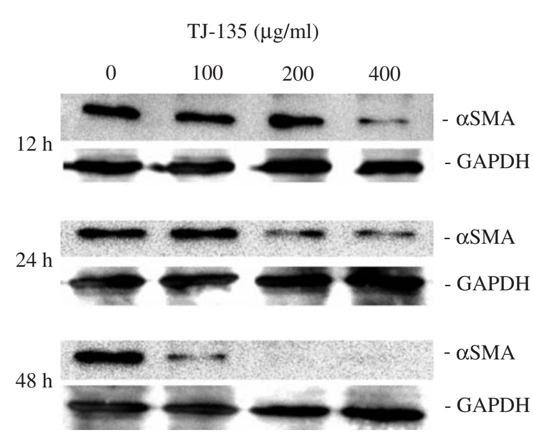


Fig. 2. Effect of TJ-135 on α-smooth muscle actin expression in HSCs. HSCs were cultured with various concentrations of TJ-135 for various periods of time. Then, α-smooth muscle actin expression in the cells was determined by immunoblotting. GAPDH expression was also analyzed as a control for protein loading. Representative immunoblot of three experiments is shown.

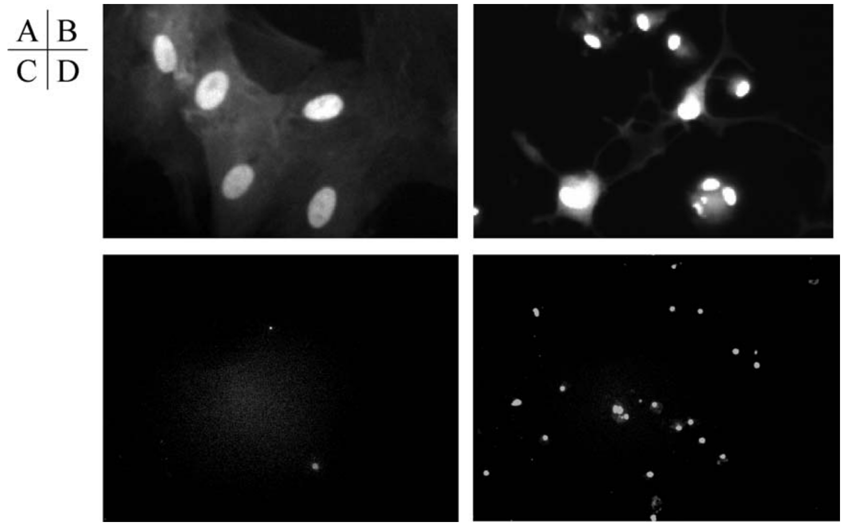


Fig. 3. Effect of TJ-135 on nuclear morphology in HSCs. HSCs were cultured with (B and D) or without (A and C) TJ-135 at 400 μg/ml for 72 h. (A and B) Nuclear morphology was assessed by Hoechst 33258 staining (×400). (C and D) HSCs containing fragmented DNA were examined by TUNEL technique (×100). Representative results of four experiments are shown.

*Effect of TJ-135 on HSC nuclear morphology, histone-associated DNA fragmentation and caspase 3 activity*

Changes in cellular morphology are important events in apoptotic cell death (Cohen, 1993; Kerr et al., 1972). Cellular shrinking was first described as one of the marked events of apoptosis (Kerr et al., 1972) and the disruption of F-actin with alteration of cytoskeleton was found in apoptosis of human ovarian cancer cells (Lee et al., 2001). Thus, we wondered whether the alteration of morphology and cytoskeletal organization by TJ-135 in HSCs might accompany apoptosis. First, the nuclear morphology of HSCs was assessed by Hoechst 33258 dye staining after 72-h treatment with 400 μg/ml TJ-135. As shown in Fig. 3, the condensation of nuclear chromatin appeared to be found more frequently in HSCs treated with TJ-135 (Fig. 3B) compared to untreated control cells (Fig. 3A). Then, the fragmentation of DNA into oligonucleosomal lengths, a further feature of apoptosis, was examined by TUNEL technique. HSCs cultured on glass chamber slides and exposed to 400 μg/ml TJ-135 for 72 h.

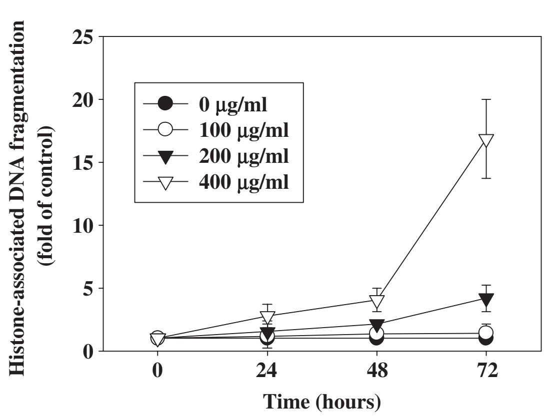


Fig. 4. Effect of TJ-135 on histone-associated DNA fragmentation in HSCs. HSCs were cultured with various concentrations of TJ-135 for various periods of time. Histone-associated DNA fragmentation was determined using a cell death detection enzyme-linked immunosorbent assay kit. Each value represents the mean ± S.D. of three experiments.

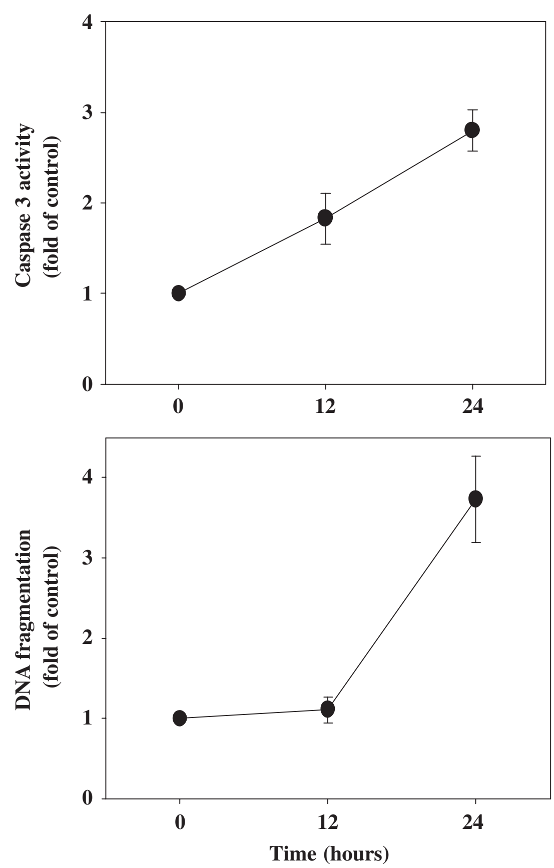


Fig. 5. Effect of TJ-135 on caspase 3 activity and histone-associated DNA fragmentation in HSCs. HSCs were cultured with 400 μg/ml TJ-135 for 12 and 24 h. Then, caspase 3 activity (upper) or histone-associated DNA fragmentation (lower) was determined using a colorimetric CaspACE kit or a cell death detection enzyme-linked immunosorbent assay kit. Each value represents the mean ± S.D. of three experiments.

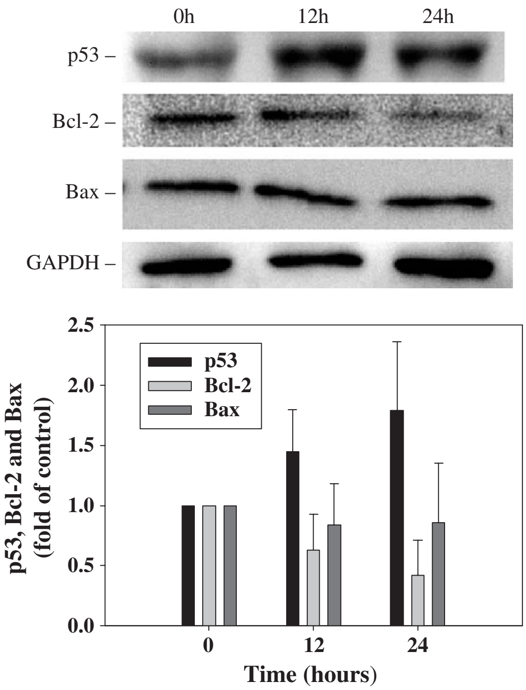


Fig. 6. Effect of TJ-135 on p53, Bcl-2 or Bax expression in HSCs. HSCs were cultured with 400 μg/ml TJ-135 for 12 and 24 h. Then, p53, Bcl-2 or Bax expression in the cells was determined by immunoblotting. GAPDH expression was also analyzed as a control for protein loading. Representative immunoblot and quantitative results of amounts of p53, Bcl-2 or Bax are shown, and columns and bars represent the means ± S.D. of four experiments.

This treatment increased the number of cells containing fragmented DNA (Fig. 3D) compared with untreated cells (Fig. 3C). Finally, we next quantitatively determined cytoplasmic histone-associated DNA oligonucleosome fragments associated with apoptotic cell death. The treatment of HSCs with TJ-135 enhanced histone-associated DNA fragmentation in a concentration-related manner as demonstrated in Fig. 4. TJ-135 at 400 μg/ml significantly induced an increase of DNA fragmentation as early as 24 h after its addition to the culture medium of HSCs.

We then investigated whether caspase 3, a key enzyme in various forms of apoptosis (Fernandes-Alnemri et al., 1994), was involved in the increased DNA fragmentation in HSC induced by TJ-135. As shown in Fig. 5, 12-h treatment with 400 μg/ml TJ-135 resulted in a two-fold increase in caspase 3 activity, and 24-h treatment, three-fold. On the other hand, the significant enhancement of DNA fragmentation by TJ-135 at 400 μg/ml was not detected at 12 h after its addition, but detected at 24 h (Fig. 5); caspase 3 activation preceded enhancement of DNA fragmentation. These findings suggest that HSC apoptosis induced by TJ-135 is mediated by caspase 3 activation.

### *Effect of TJ-135 on p53, Bcl-2 or Bax expression in HSCs*

The p53 tumor suppressor gene is crucial in some forms of apoptosis (Yonish-Rouach et al., 1991). Thus, we examined whether p53 protein level was altered in HSCs by TJ-135. Fig. 6 shows that the treatment of HSCs with 400 μg/ml TJ-135 enhanced the p53 protein level in HSCs.

Next, possible mechanism by which TJ-135 induces HSC apoptosis was examined by quantifying any associated changes in cellular content of proteins that inhibit or enhance apoptosis, that is Bcl-2 (Hockenbery et al., 1991) or Bax (Oltvai et al., 1993). As shown in Fig. 6, immunoblot analysis of Bcl-2 or Bax extracted from HSCs showed that the treatment with 400 μg/ml TJ-135 reduced the Bcl-2 protein level, but not Bax.

### *Effect of TJ-135 on Akt or Bad expression in HSCs*

Akt is considered a key factor for cell proliferation and apoptosis (Chan et al., 1999; Dudek et al., 1997) and is shown to be abundantly expressed in culture-activated HSCs (Fischer et al., 2001). In those cells, addition of proapoptotic agent, peripheral-type benzodiazepine receptor ligand, decreased the level of phosphorylated Akt (Fischer et al., 2001). Thus, we examined if TJ-135 might alter Akt protein level and its phosphorylation in HSCs. As demonstrated in Fig. 7, the treatment of HSCs with 400 μg/ml TJ-135 reduced the level of phosphorylated Akt as early as 30 min without affecting Akt protein level.

Phosphorylation of Akt was shown to exert anti-apoptotic effects as a result of the phosphorylation of its substrate Bad (Chan et al., 1999). In HSC apoptosis, Bad is shown to be

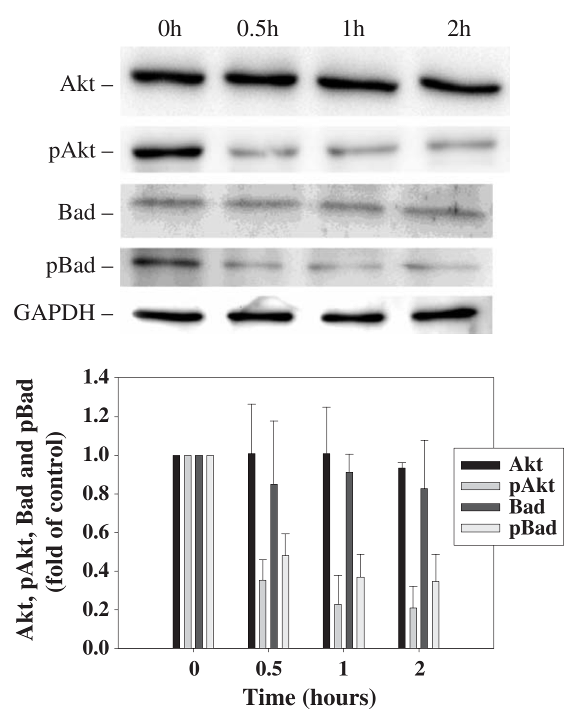


Fig. 7. Effect of TJ-135 on Akt or Bad expression in HSCs. HSCs were cultured with 400 μg/ ml TJ-135 for various periods of time up to 2 h. Then, Akt, phosphorylated Akt (pAkt) Bad or phosphorylated Bad (pBad) expression in the cells was determined by immunoblotting. GAPDH expression was also analyzed as a control for protein loading. Representative immunoblot and quantitative results of amounts of Akt, pAkt, Bad or pBad are shown, and columns and bars represent the means ± S.D. of four experiments.

involved in the Akt-mediated signaling pathway (Fischer et al., 2001). Thus, we next determined the possible associated changes in cellular content of Bad and phosphorylated Bad. Fig. 7 showed that the treatment of HSCs with 400 μg/ml TJ-135 reduced the level of phosphorylated Bad without affecting Bad protein level.

## Discussion

TJ-135 is an aqueous extract from three herbs; *Artemisia capillaries spica*, *Gardenia fructus* and *Rhei rhizome*. It has been known that *A. capillaries spica* acts as choleretic, and *G. fructus* promotes bile secretion and *R. rhizome* acts as a laxative (Miwa, 1953; Komiya et al., 1976). In the current study, we first found the morphological alteration of shrinking with fewer stress fibers of HSCs caused by this TJ-135, which are often associated with apoptotic process (Lee et al., 2001; Kerr et al., 1972). Then, we detected enhanced condensation of nuclear chromatin by TJ-135 and determined an increase in caspase 3 activity and histone-associated DNA fragmentation. These evidence of increased caspase 3 activity and DNA fragmentation with the morphological change often found in apoptosis (Lee et al., 2001; Kerr et al., 1972) suggests that TJ-135 induces HSC apoptosis in culture.

We also determined the possible modulation of pro- and anti-apoptotic proteins in the treatment of HSCs with TJ-135 and found that enhanced expression of pro-apoptotic protein, p53 (Yonish-Rouach et al., 1991), and reduced expression of anti-apoptotic proteins, Bcl-2 (Hockenbery et al., 1991), phosphorylated Akt (Chan et al., 1999; Dudek et al., 1997) and phosphorylated Bad (Chan et al., 1999). These alterations are consistent with the induction of apoptosis in HSCs by TJ-135 and it is speculated that these pro- and anti-apoptotic proteins might play some role in the effect of TJ-135. Regarding Akt, Imanishi et al. reported that TJ-135 reduced its phosphorylation in HSCs in culture (Imanishi et al., 2004), where they speculated that the alteration of the phosphorylation of Akt might contribute to the inhibition of cell migration. In addition to this finding, we found that TJ-135 reduced the level of phosphorylated Bad in HSCs as described above. Thus, Akt might play a role in another action, apoptosis, exerted by TJ-135 in HSCs.

It has been reported that TJ-135 at 100 μg/ml suppressed proliferation and fibrogenesis in cultured rat HSCs (Imanishi et al., 2004) and in cultured human HSCs (Sakaida et al., 2003). In this study, we found that TJ-135 at higher concentration, 400 μg/ml, induces apoptosis in cultured rat HSCs. It should be noted that 500 μg/ml TJ-135 was shown to suppress apoptosis in hepatocytes (Yamamoto et al., 1996). Thus, TJ-135 at 400 μg/ml is not generally toxic in various cells. Because TJ-135 is a mixture extracted from three herbs and could be converted in vivo to metabolites by the bacterial gut flora before exerting their effects, the clinical relevance of TJ-135 concentrations used in this study, 200 μg/ml and 400 μg/ml, may be difficult to be determined. It might be noted that TJ-135 has been reported to suppress liver injury in mice induced by Fas (Yamamoto et al., 2000) and its mechanism is explained based on the finding that TJ-135 at 500 μg/ml suppresses apoptosis in hepatocytes in vitro (Yamamoto et al., 1996). Similarly, sho-saiko-to, another herbal medicine used in the same dosage as TJ-135 for the clinical patients, suppresses hepatic fibrosis (Shimizu et al., 1999; Sakaida et al., 1998) and its mechanism is explained by the evidence that sho-saiko-to at 500 μg/ml and 1000 μg/ml inhibits collagen production and proliferation in cultured rat HSCs (Sakaida et al., 2004; Kayano et al., 1998). Because TJ-135 concentrations used in this study are less than those in these in vitro studies, they might also have some relevance to the effect in vivo.

The inhibition of actin polymerization was reported to suppress apoptotic body formation in cells undergoing apoptosis (Levee et al., 1996; Cotter et al., 1992). On the other hand, damage to cytoskeletons, as exemplified by actin disruption (Rubtsova et al., 1998; Van de Water et al., 1996) and microtubule disassembly (Torres and Horwitz, 1998; Blagosklonny et al., 1997), often causes apoptosis in various cells. Hence, it is suggested that there are many close interactions between the state of cytoskeletons and progress in apoptosis (Yamazaki et al., 2000). In this study, the treatment of HSCs with TJ-135 clearly caused disruption of actin cytoskeleton and apoptosis, suggesting that our finding might be one of the latter instances that damages to cytoskeleton causes apoptosis. Moreover, we found that TJ-135 reduced the expression of α-smooth muscle actin. Of interest is the report that α-smooth muscle actin is cleaved by caspase 3 in the process of apoptosis (Goldman et al., 2003; Nakazono-Kusaba et al., 2002). Whether the integrity of actin cytoskeleton including α-smooth muscle actin might determine the fate of activated HSCs should be further investigated. As to the relation between the cellular proteins involved in the mechanism of apoptosis and the state of cytoskeletons, disruption of actin microfilaments by cytochalasin D was shown to lead to activation of p53 in fibroblasts (Rubtsova et al., 1998), and Bcl-2 was reported to keep microtubule integrity (Haldar et al., 1997). Thus, our finding is in line with these reports suggesting the link between alteration of cytoskeleton and p53 or Bcl-2. The question whether the alteration of cytoskeletons would directly affect pro- or anti-apoptotic proteins or whether caspase would play some role in disruption of cytoskeletons should be clarified in the future.

It is now well known that HSCs play a central role in the development of hepatic fibrosis (Friedman, 1993). In response to liver damage, HSCs “activate” to a myofibroblast-like phenotype with enhanced proliferation, fibrogenesis and contractility (Rockey et al., 1993; Minato et al., 1983; Okanoue et al., 1983). On the other hand, current evidence demonstrates that the mechanisms of spontaneous resolution of rat hepatic fibrosis involve HSC apoptosis, suggesting that HSCs may also play a pivotal role in the resolution of hepatic fibrosis (Issa et al., 2001; Iredale et al., 1998). Moreover, gliotoxin which induces HSC apoptosis has been shown to reduce experimental hepatic fibrosis in rats (Wright et al., 2001). Thus, the induction of HSC apoptosis could be involved in the mechanism how TJ-

135 works on resolution of hepatic fibrosis. Our current data may shed light on the novel effect of herbal medicine, which is gaining increasing attention in the field of liver diseases (Bean, 2002).

## Acknowledgment

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